

Unit-4

Multimodal Applications

MULTIMODAL MACHINE LEARNING

MULTIMODAL AI

Bridging Language, Vision & Sound



The Modality Spectrum

- **Text:** Discrete tokens, semantic richness (NLP).
- **Audio:** Waveforms, spectrograms, prosody (Speech).
- **Image/Video:** Spatial and temporal pixel correlations.
- **Sensors:** LiDAR, IMU, GPS, Heat maps (Robotics/IoT).

Representation & Alignment

- **Joint Representations:** Mapping all modalities into a single shared vector space.
- **Coordinated Representations:** Keeping modalities separate but enforcing similarity (e.g., Contrastive Learning).
- **Alignment:** Finding the direct correspondence between sub-elements (e.g., which word matches which video frame?).

Representation & Alignment

- Alignment is perhaps the hardest part of multimodal AI. It is categorized into two types:
 - **Explicit Alignment:** Directly matching parts of the data. For example, in **Visual Grounding**, we align the phrase "the red ball" with the specific bounding box in an image.
 - **Implicit Alignment:** The model learns to attend to relevant parts of one modality while processing another (e.g., **Attention Mechanisms** in Transformers where the text query "looks" at specific image patches).

Co-learning & Modality Relationships

- **Co-learning:** Transferring knowledge from one modality to help another.
- **Complementary:** Modalities provide different info (e.g., Text says "happy," Video shows a smile).
- **Redundant:** Modalities provide the same info (e.g., Audio "Stop" and a red light).

Co-learning (The "Teacher-Student" Paradigm)

- Co-learning allows a model to learn a task in one modality by leveraging the labels or features of another.
- **Example:** You have a massive dataset of videos with audio, but only a small dataset of just audio. You can use the visual information to help the model learn better audio features (cross-modal transfer).

Complementary vs. Redundant

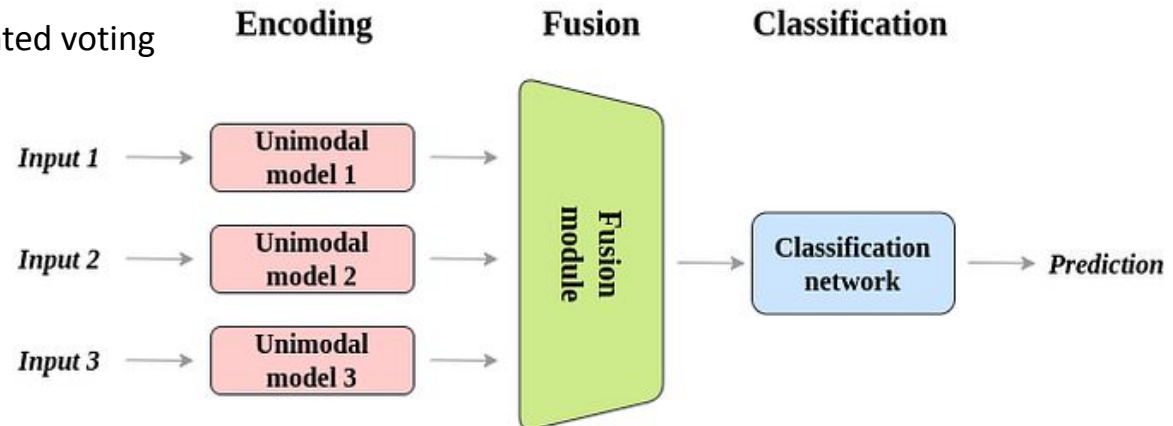
- **Redundancy** increases **reliability**. If the microphone is noisy, the visual lip-reading (redundant signal) helps the model stay accurate.
- **Complementarity** increases **context**. A video of a person crying could be "sadness" or "joy." The audio (sobbing vs. laughing) provides the complementary info needed to disambiguate.

The "Nightmare" Scenarios (Challenges)

- Missing Modality (The "Broken Sensor" problem).
- Modality Imbalance (One signal dominates the others).
- Temporal Alignment (The "Lag" problem).
- Noisy Modalities (Low-quality or corrupted inputs).

Model-Free Fusion Approaches

- Early Fusion
 - Image_features + Audio_features → Neural Network
- Intermediate Fusion
 - Image model → Embedding
 - Audio model → Embedding
 - Joint embedding → Neural Network
- Late Fusion
 - Image model → prediction
 - Audio model → prediction
 - Final output = weighted voting



Hybrid Fusion

- **1. Hierarchical (Multi-Stage) Hybrid Fusion**

- This is the most common form. It performs fusion at multiple levels of the neural network.
- **The Workflow:** Local features from two modalities (e.g., Audio and Video) are fused early on to create "mid-level" features. These mid-level features are then processed alongside a third modality (e.g., Text) at a later stage.

- **2. Parallel (Joint-Decision) Hybrid Fusion**

- In this setup, the model runs an Early Fusion branch and a Late Fusion branch simultaneously.
- **The Workflow:** 1. Branch A: Concatenates features at the input level to learn joint correlations. 2. Branch B: Processes modalities independently to maintain the integrity of each signal. 3. The Hybrid Step: The outputs of Branch A and Branch B are merged (usually via a weighted average or a meta-classifier) to make the final prediction.

- **3. Gated or Adaptive Hybrid Fusion**

- This is a more "intelligent" version of hybrid fusion that uses an internal mechanism to decide **when** and **how much** to fuse.
- **The Workflow:** It uses **Gated Neural Networks** (similar to LSTMs or GRUs). A "Controller" or "Attention" sub-network looks at the input quality. If the Audio is noisy, the gate suppresses the early-fusion features and relies more on the Late-fusion visual features.

Kernel Based Fusion

- Most multimodal data is not linearly separable in its raw form. For example, you cannot draw a straight line between "Happy" and "Sad" based on raw pixel values and audio frequencies.
- **The Math:** A kernel function $k(x, y)$ calculates the inner product of two data points in a high-dimensional feature space without ever explicitly calculating the coordinates in that space.
- **Multimodal Application:** We define a specific kernel for each modality:

Multiple Kernel Learning (MKL)

- MKL is the advanced version of kernel fusion. Instead of picking one "perfect" kernel, we use a **weighted combination** of several kernels.
- **The Fusion Equation**
 - The fused kernel K_{fused} is defined as:

$$K_{fused}(x, y) = \sum_{m=1}^M \beta_m K_m(x_m, y_m)$$

Types

- **A. Early Kernel Fusion (Composite Kernels)**
 - We combine the data *inside* the kernel function.
 - **Pro:** Captures cross-modal correlations.
 - **Con:** Requires all modalities to be present and synchronized.
- **B. Intermediate Kernel Fusion (MKL)**
 - This is the most popular type. We compute separate kernel matrices for each modality and then combine the matrices.
 - **Pro:** You can use different types of kernels for different data (e.g., a Histogram Intersection kernel for images and a Linear kernel for text).
- **C. Late Kernel Fusion**
 - We train separate SVMs (Support Vector Machines) for each modality kernel and fuse their output scores (similar to Late Fusion in Neural Networks).

Factorial Hidden Markov Models

- Imagine you are monitoring a person's health using a **Biometric System** (one of your case studies). You have:
- **Chain 1 (Hidden):** The person's actual physical exertion level (Resting, Walking, Running).
- **Chain 2 (Hidden):** The person's emotional state (Calm, Stressed, Excited).
- Both of these "hidden" chains exist simultaneously. You don't see them directly. Instead, you observe **one combined signal**: a heart rate monitor (Observation).

Mathematical Structure

2. Mathematical Structure

In a standard HMM, we have one state q_t . In an FHMM with M layers, the state at time t is a collection of M states:

$$S_t = \{s_t^{(1)}, s_t^{(2)}, \dots, s_t^{(M)}\}$$

The Three Key Components:

1. **Independent Transitions:** Each chain m evolves independently according to its own transition matrix $P(s_t^{(m)} | s_{t-1}^{(m)})$.
2. **Joint Observation:** The observation O_t depends on the current state of **all** chains:
 $P(O_t | s_t^{(1)}, \dots, s_t^{(M)})$.
3. **Factorization:** The "Factorial" name comes from the fact that the state space is the Cartesian product of the individual chains' state spaces.

Why it's Crucial for Multimodal Fusion

- FHMMs are the gold standard for **Source Separation** (The "Cocktail Party Problem").
- **Example:** You have one microphone (Observation) recording two people talking (Two Hidden Chains).
- **The Problem:** A standard HMM would need $N \times M$ states to model this, which leads to "State Explosion."
- **The FHMM Solution:** It models them as two separate processes that happen to be heard by one sensor.

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THANK YOU

